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Hydrothermal synthesis and optimization of boron doped $\text{LiZr}_2(\text{PO}_4)_3$ Li-ion solid electrolyte

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ABSTRACT

Boron doped $\text{LiZr}_2(\text{PO}_4)_3$ electrolyte was successfully synthesized via facile hydrothermal synthesis. Our results show that boron doping mainly happens on the Zr-site of $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$ not the P-site of $\text{Li}_{1+6x}\text{Zr}_2(\text{P}_{1-x}\text{B}_x\text{O}_4)_3$. Rietveld refinement of the unit-cell parameters was performed, and it was verified by consideration of Vegard's law that it is possible to obtain phase purity up to $x = 0.05$ in $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$. This corresponds with phases present in the XRD data, which shows the additional presence of the low temperature (monoclinic) phase for the powder sintered at 1200 °C for 12 h of compositions with $x \geq 0.075$. The compositions inside the solid solution undergo the phase transition from triclinic to rhombohedral when heating from 25 to 100 °C.

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1. Introduction

The coating of the lithium metal anode in Li-air batteries by water stable lithium ion conducting solid electrolytes, as separators, has been suggested to prevent damage by atmospheric moisture and oxygen [1–3]. Such lithium-ion conducting solid electrolytes would also find application in all-solid-state Li-ion batteries. These solid materials ideally should offer Li-ion conduction greater than $10^{-4} \text{ S cm}^{-1}$ at room temperature [4]. Nowadays, the best commercially available solid Li-electrolyte is that of $\text{Li}_{1.3}\text{Ti}_{1.7}\text{Al}_{0.3}(\text{PO}_4)_3$ [1–5], having a similar crystallographic structure to that of rhombohedral sodium super ionic conductors (NASICON), from the sodium zirconium phosphate (NZP) structural family [3,4]. However, the presence of the reducible ion Ti in this material, limits its electrochemical performance. Very recently, advances have been offered by the study of the material $\text{Li}_{1.2}\text{Zr}_{1.9}\text{Ca}_{0.1}(\text{PO}_4)_3$

where the zirconium based phosphate offers a much higher electrochemical stability.

The general formula of the NZP materials is given by $\text{M}_x\text{A}_2(\text{TO}_4)_3$. Positions T commonly represent phosphate groups, although a wide range of further elements have also been reported such as, arsenic, antimony, silicon, bismuth, boron, molybdenum and others [5–7]. The essential requirement is for these elements to form tetrahedral, TO_4 , groups. Atoms A occupy octahedral AO_6 arrays and are predominantly trivalent or tetravalent cations. Mixtures (III–IV) or partial substitution of these elements by di- or pentavalent cations have also been documented [6–8]. These tetrahedral, TO_4 , and octahedral AO_6 building blocks form the framework of the NZP materials, while the resultant interstitial sites are filled by M-cations. The M-cations are commonly ions of unit charge, such as alkali metal, hydrogen, ammonium groups.

This highlights the vast compositional flexibility of this structural family and, hence, an almost unrivalled capacity with which to tailor properties. Indeed, fast Na-ionic conduction has been introduced into these materials by adoption of suitable substitution mechanisms, such as $\text{Na}_{1+3x}\text{Zr}_2(\text{P}_{1-x}\text{Si}_x\text{O}_4)_3$ (NASICON) where peak performance is offered in the range $1.8 < x < 2.3$ [9].

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Conversely, the end members of this series $\text{NaZr}_2(\text{PO}_4)_3$ and $\text{Na}_4\text{Zr}_2(\text{SiO}_4)_3$ offer poor ionic conduction. In the former case, Na is confined to the Mi sites due to the narrowness of the Mi to Mii passageways, whereas in the latter all Mii and Mii sites are fully occupied. It is only by the correct dopant strategy and removal of steric hindrances that the Mii sites can be partial filled and ion migration can occur. The lower ionic radii of Li allows it to reside in both Mi and Mii sites in the composition lithium zirconium phosphate, $\text{LiZr}_2(\text{PO}_4)_3$, leading to higher conduction than the Na analogue [10–12]. Further improvements can be made by adopting suitable solid solution mechanisms such as $\text{Li}_{3+2x}(\text{Sc}_{1-x}\text{Mg}_x)_2(\text{PO}_4)_3$, $\text{Li}_{3-2x}(\text{Sc}_{1-x}\text{Ti}_x)_2(\text{PO}_4)_3$ and $\text{Li}_{3-4x}(\text{Sc}_{1-x}\text{Nb}_x)_2(\text{PO}_3)_2$ [13]. Note similar to the Na-case, such improvements in conductivity relate to improved distribution of Li over the Mi and Mii sites and partial occupancy. It is only when these criteria are met that peak fast ion conductivity can be obtained. However, they can also be partially or fully occupied by ions of higher charges ($2+$ or $3+$) or even absent [6,7]. There are four such interstitial sites per formula unit, described by the crystallographic formula $(\text{Mi})(\text{Mii})_3\text{A}_2(\text{TO}_4)_3$, where Mi and Mii have different crystallographic orientations, connected by narrow passageways. In the current work, improvements have been made in the design of lithium ion conducting solid electrolytes as separators in Li-air batteries by attempting to dope with boron (B) on the T-site, $\text{Li}_{1+6x}\text{Zr}_2(\text{P}_{1-x}\text{B}_x\text{O}_4)_3$, and A-site, $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$. This aims to both increase the Li-content and to distribute the Li over the two distinct interstitial crystallographic sites, as needed for fast ionic motion. Boron (B) was reported to have been successfully doped in NZP materials both on the A-site [14] (called here Zr site for $\text{LiZr}_2(\text{PO}_4)_3$) and the T-site [15] (called here P site for $\text{LiZr}_2(\text{PO}_4)_3$), showing that B can potentially go on either the Zr-site or P-Site of $\text{LiZr}_2(\text{PO}_4)_3$. $\text{LiZr}_2(\text{PO}_4)_3$ has been reported to be successfully synthesized by conventional solid state reaction [4,16] and sol-gel methods [16,17]. With the samples prepared by conventional solid state reaction undergoing repeated thermal treatments at 1100°C for 16 h and above 1300°C for around 16 h [16], increasing the risk of lithium volatilization. In contrast, the samples prepared by hydrothermal processes undergo thermal treatments of at maximum 1200°C for 12 h [16], decreasing the risk of lithium volatilization when compared with the solid state reaction. $\text{LiZr}_2(\text{PO}_4)_3$ can either be sintered at a so called high temperature of 1200°C , crystallizing in a triclinic form at room temperature or at a so called low temperature of 900°C , crystallizing in a monoclinic form at room temperature. Conversely, at temperatures between 900 and 1200°C $\text{LiZr}_2(\text{PO}_4)_3$ presents a mixture of both triclinic and monoclinic phases [10,11]. The high temperature triclinic phase suffers a phase transition to a rhombohedral form at around 40 – 60°C [10,18]. The high temperature phase offers a sharp increase in lithium mobility leading to high ionic conductivity above the phase transition from triclinic to rhombohedral with $1.2 \times 10^{-2} \text{ S cm}^{-1}$ at 300°C , while the low temperature phase has more hindered lithium motion with conductivity limited to values around $5 \times 10^{-4} \text{ S cm}^{-1}$ at 300°C [18].

Thus, to minimize lithium losses, it was decided to synthesize $\text{LiZr}_2(\text{PO}_4)_3$ using a hydrothermal method and to start by trying to dope $\text{LiZr}_2(\text{PO}_4)_3$ with boron on the P-site due to the more similar ionic radii of B^{3+} and P^{5+} , $B = 0.25 \text{ \AA}$ Vs $P = 0.31 \text{ \AA}$ [19], useful for forming the 4-coordinate tetrahedron necessary for the P-site. Secondly, attempts were made to dope $\text{LiZr}_2(\text{PO}_4)_3$ with B on the Zr-site. Here B^{3+} offers a much smaller ionic radius than Zr^{4+} , $B = 0.41 \text{ \AA}$ Vs $\text{Zr} = 0.86 \text{ \AA}$ [19], when forming the 6-coordinate octahedron as required for the Zr-site. Due to the higher ionic conduction of the high temperature phase, it was also decided to sinter the high temperature phase instead of the low temperature phase, but attempting from the beginning to avoid temperatures as high as 1200°C .

2. Experimental procedure

2.1. Synthesis of $\text{Li}_{1+6x}\text{Zr}_2(\text{P}_{1-x}\text{B}_x\text{O}_4)_3$ and of $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$ ($x = 0$ – 0.25)

Lithium zirconium phosphate, $\text{LiZr}_2(\text{PO}_4)_3$, with $x = 0$ and 0.15 , powders were synthesized based on a hydrothermal method adapted from a recent work by Kim et al. [17]. The compositions doped with boron on the P-site [$\text{Li}_{1+6x}\text{Zr}_2(\text{P}_{1-x}\text{B}_x\text{O}_4)_3$], were prepared by mixing stoichiometric amounts of the precursors orthophosphoric acid, H_3PO_4 , (Carlo Erba 85% purity); zirconium (IV) acetate hydroxide, $(\text{CH}_3\text{COO})_x\text{Zr}(\text{OH})_y$, with $x + y = 4$, (Sigma-Aldrich 100% purity); boric acid, H_3BO_3 , (Merck 99.8% purity); lithium acetate dehydrate, $\text{CH}_3\text{COOLi} \cdot 2\text{H}_2\text{O}$, (Sigma-Aldrich 97% purity); and using citric acid (CA), $\text{C}_6\text{H}_8\text{O}_7$, (Sigma-Aldrich 99.5% purity), as the chelating agent. Lithium zirconium phosphate doped with boron on the Zr-site [$\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$] was prepared by mixing stoichiometric amounts of the same precursors used to prepare $\text{Li}_{1+6x}\text{Zr}_2(\text{P}_{1-x}\text{B}_x\text{O}_4)_3$.

H_3PO_4 , $(\text{CH}_3\text{COO})_x\text{Zr}(\text{OH})_y$, H_3BO_3 , and $\text{CH}_3\text{COOLi} \cdot 2\text{H}_2\text{O}$ were dissolved in deionised water, in that respective order. Citric acid was then added, in a molar ratio CA/M: 7.5/1 ($M = [\text{Li}] + [\text{Zr}]$) adapted from Barre et al. [20]. The B doping was attempted by introducing H_3BO_3 in the solution, as mentioned above. The solution was continuously heated at 180°C in oven for 12 h. The resulting filtered powder was fired in air at 750°C for 5 h, with heating and cooling rates of $1^\circ\text{C}/\text{min}$, in order to burn the organics present. The resulting powder was ground with a mortar and pestle and pressed uniaxially into pellets (8 mm in diameter) for 1 min at 20 MPa, followed by isostatic pressing for 15 min at 450 MPa. The pellets were fired using sacrificial powder of the same composition, following a two-step sintering cycle, with a peak temperature of 1000°C for 30 min and a dwell temperature of 900°C for 10 h, with heating to 1000°C at $5^\circ\text{C}/\text{min}$, cooling to 900°C at $10^\circ\text{C}/\text{min}$ and cooling to 20°C at $5^\circ\text{C}/\text{min}$. Both the calcined and sintered powders were analyzed by X-ray diffraction (XRD) using a Rigaku Geigerflex diffractometer (Cu K radiation, 10 – $80^\circ 2\theta$, step width 0.02° , scan rate $3^\circ/\text{min}$).

The oven dried powder of $\text{Li}_{1.20}\text{Zr}_{1.80}\text{B}_{0.20}(\text{PO}_4)_3$ was analyzed by TGA using a Netzsch STA 449F3 TG-DTA/DSC instrument in flowing air. Thermal expansion or contraction behaviour in air was followed using a Linseis dilatometer on a green compact rectangular sample of $\text{Li}_{1.025}\text{Zr}_{1.975}\text{B}_{0.025}(\text{PO}_4)_3$. Microstructural analysis was performed in sintered samples of $\text{LiZr}_2(\text{PO}_4)_3$ and $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$ by scanning electron microscopy (SEM) using a Hitachi S4100 equipment. Rietveld refinement of the unit cell parameters, volume V , a , b and c , was performed from the XRD data of the sintered powders of $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$, with $x = 0$, 0.025 , 0.050 , 0.075 and 0.100 , using the FullProf Suite Program (2.05), to find the solid solution limit of B on the Zr-site. Sintered powders of $\text{LiZr}_2(\text{PO}_4)_3$, $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$ and $\text{Li}_{1.10}\text{Zr}_{1.90}\text{B}_{0.10}(\text{PO}_4)_3$ were analyzed by XRD as a function of temperature, on a HTK 18 N High-Temperature Chamber, using a PANalytical X'Pert Pro Diffractometer (Cu K radiation, 10 – $80^\circ 2\theta$, step width 0.02° , scan rate $100^\circ/\text{step}$), to study the evolution of the powder XRD patterns with temperature in 100°C steps, from 25 to 800°C .

2.2. Hydrostatic press and fabrication of the pellets

Discs of 8 mm diameter were obtained by uniaxial pressing at 20 MPa for 1 min. Then, the samples were sealed in vacuum in plastic bags. A hydrostatic press (Hiperbaric 55, Hiperbaric, Burgos, Spain) was utilized to press the pellets in a range of pressures 450 MPa. This equipment has a pressure vessel of 200 mm inner diameter and 2000 mm length and a maximum operation pressure of

750 MPa. It is connected to a refrigeration unit (RMA KH 40 LT, Ferrol, San Bonifacio, Italy) that allows it to control the temperature of the input water used as a pressurizing fluid. All the hydrostatic press lasted 30 min. After press, the pellets were sintered at different temperatures for further analysis.

3. Results and discussion

3.1. Sintering method selection

A TGA of the filtered powder before and after a calcination step at 350 °C, was graphically represented in Fig. 1. It can be observed that it is possible to calcine the gel directly at temperatures above 500 °C, but based on the previous work it was decided to calcine the powder at 600 °C.

3.2. Phase characterization

Fig. 2 shows the XRD results for the sintered powders of B doped $\text{LiZr}_2(\text{PO}_4)_3$ that were synthesized based on a hydrothermal method adapted from the work by Kim et al. [17]. One can observe from those XRD spectra that a good crystallinity was achieved after sintering. However, a pure $\text{LiZr}_2(\text{PO}_4)_3$ phase was not obtained and ZrP_2O_7 impurity peaks were always detected. One can also observe

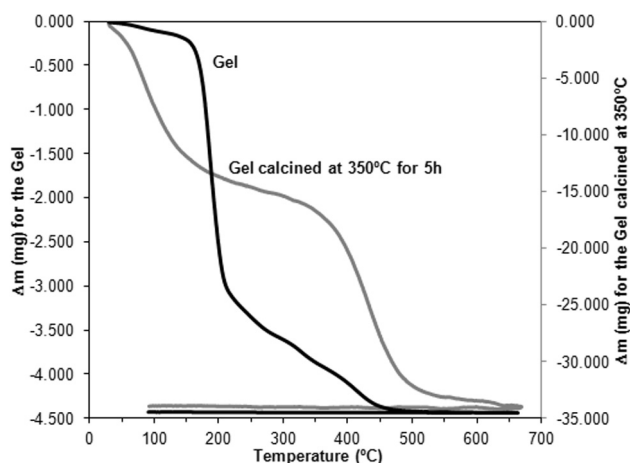


Fig. 1. TGA of the gel and the gel calcined at 350 °C for 5 h, for $\text{LiZr}_2(\text{PO}_4)_3$, for the modified method with acetic acid.

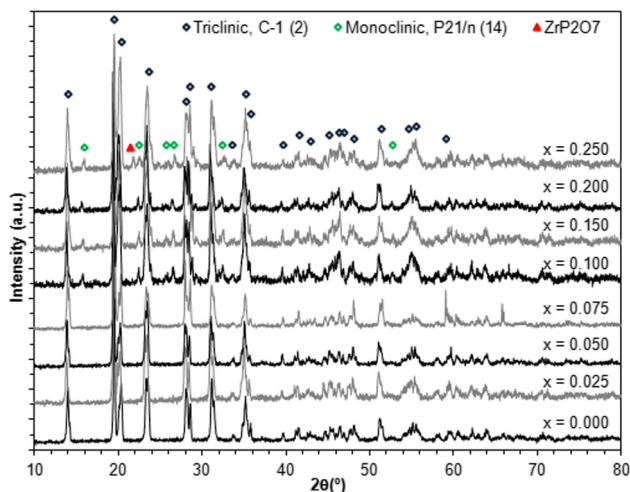


Fig. 2. XRD spectra of $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$, after sintering at 1200 °C for 12 h, with $x = 0; 0.025; 0.050; 0.075; 0.100; 0.150; 0.200$ and 0.250 from the bottom to the top.

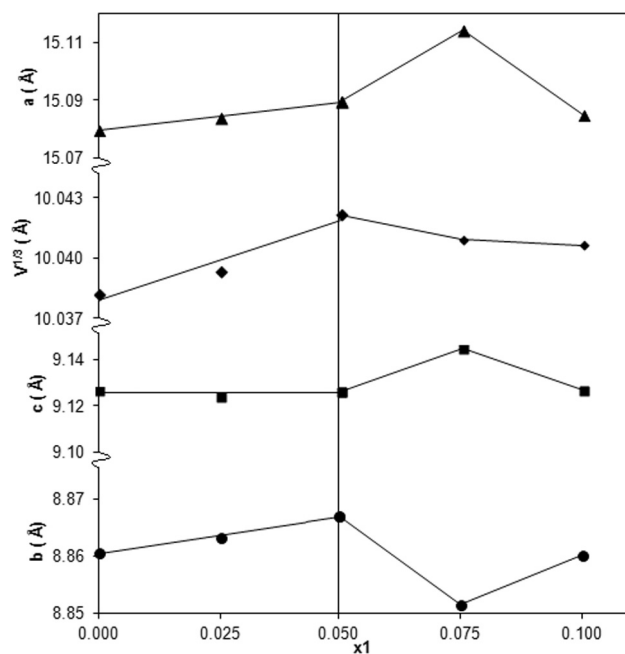


Fig. 3. Graphic representation of the cubic root of the volume ($V^{1/3}$), with a , c and b parameters of the unit-cell (the order from the top to the bottom is: $V^{1/3}$, a , c and b).

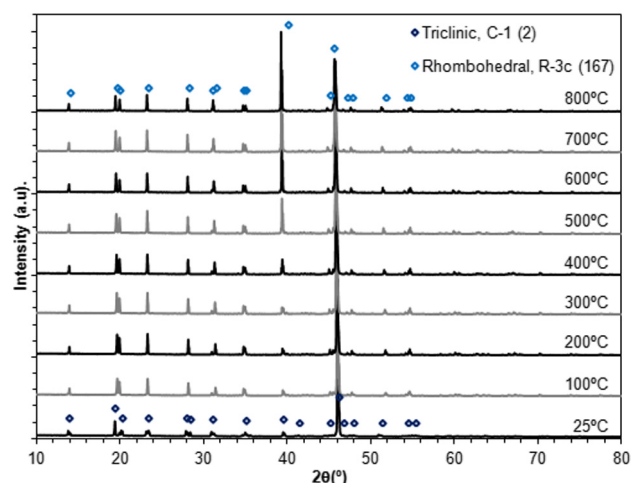


Fig. 4. Evolution of powder XRD patterns with temperature for $\text{LiZr}_2(\text{PO}_4)_3$, after sintering at 1200 °C for 12 h, from 25 to 800 °C from the bottom to the top.

that a mixture of both the high temperature phase (triclinic) and the low temperature phase (monoclinic) is present at room temperature in accordance with the bibliographic references [10,11].

From Fig. 2, one can easily observe that for $x \geq 0.10$ there is a mixture of both the high (triclinic) and the low (monoclinic) temperature phases, with impurity peaks of ZrP_2O_7 appearing for $x = 0.25$. This data suggest that the $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$ system has reached the solid solution limit at $x = 0.075$, where neither impurity phases or low temperature phase (monoclinic) peaks are observed. Further XRD data treatment by Rietveld refinement, graphically represented in Fig. 3, suggests, by assessment of Vegard's behaviour, that the solid solution limit of B in Zr site in $\text{Li}_{1+x}\text{Zr}_{2-x}\text{B}_x(\text{PO}_4)_3$ compositions lies closer to $x = 0.050$, after which there is a change in the linear dependence of the cell parameters values with B content.

To better understand these data and the structural phenomena happening with temperature, new XRD spectra were acquired at different temperatures. It is explained in the literature that both

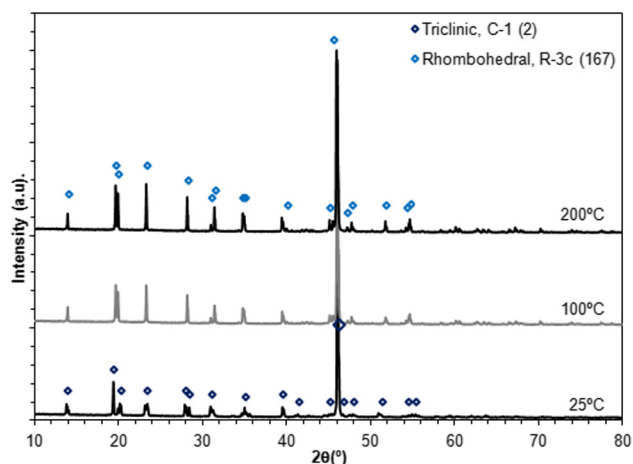


Fig. 5. Evolution of powder XRD patterns with temperature for $\text{LiZr}_2(\text{PO}_4)_3$, after sintering at 1200 °C for 12 h, for 25, 100 and 200 °C from the bottom to the top.

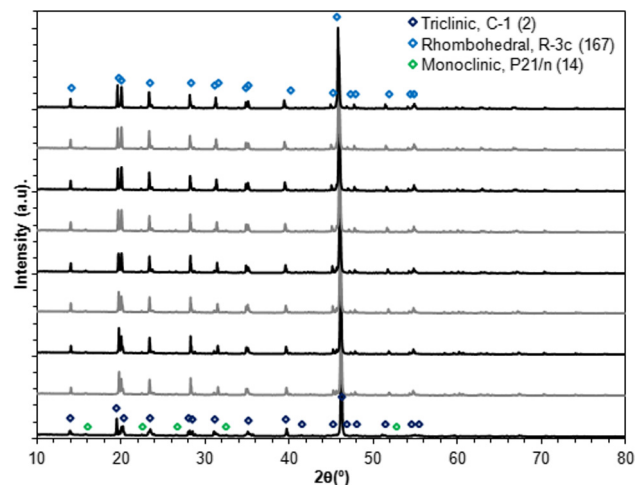


Fig. 8. Evolution of powder XRD patterns with temperature for $\text{Li}_{1.10}\text{Zr}_{1.90}\text{B}_{0.10}(\text{PO}_4)_3$, after sintering at 1200 °C for 12 h, 25 to 800 °C from the bottom to the top.

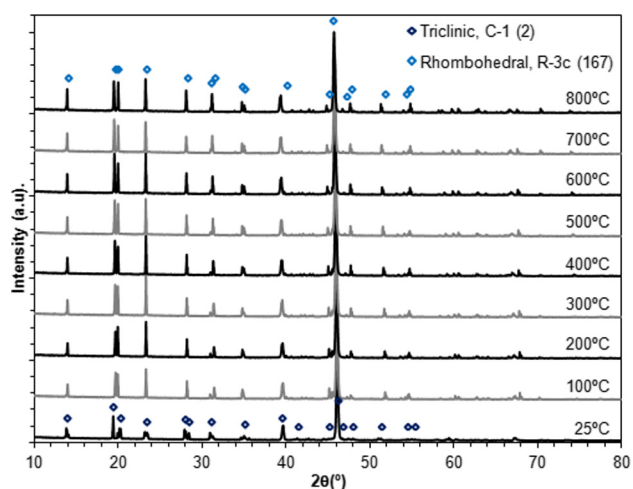


Fig. 6. Evolution of powder XRD patterns with temperature for $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$, after sintering at 1200 °C for 12 h, from 25 to 800 °C from the bottom to the top.

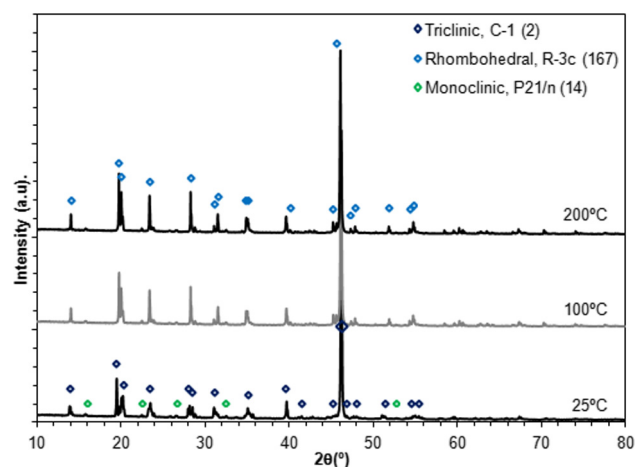


Fig. 9. Evolution of powder XRD patterns with temperature for $\text{Li}_{1.10}\text{Zr}_{1.90}\text{B}_{0.10}(\text{PO}_4)_3$, after sintering at 1200 °C for 12 h, 25, 100 and 200 °C from the bottom to the top.

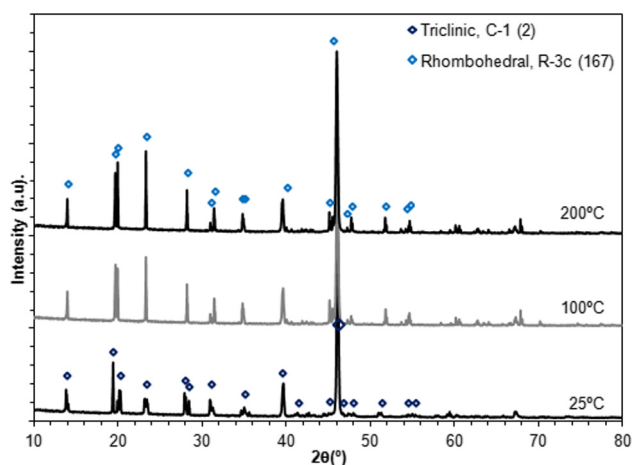


Fig. 7. Evolution of powder XRD patterns with temperature for $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$, after sintering at 1200 °C for 12 h, 25, 100 and 200 °C from the bottom to the top.

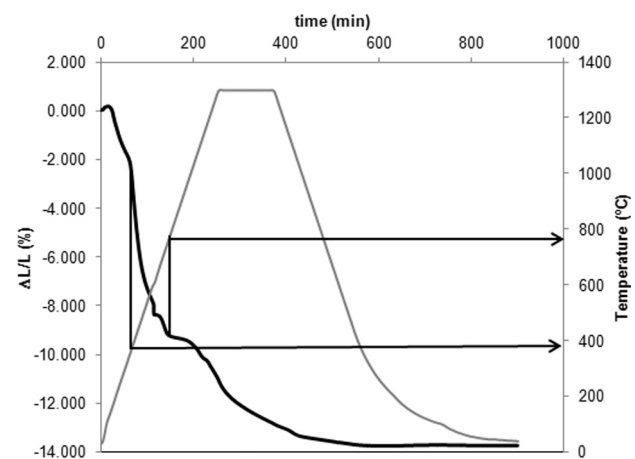


Fig. 10. Dilatometry of a green compact rectangular sample of $\text{Li}_{1.025}\text{Zr}_{1.975}\text{B}_{0.025}(\text{PO}_4)_3$.

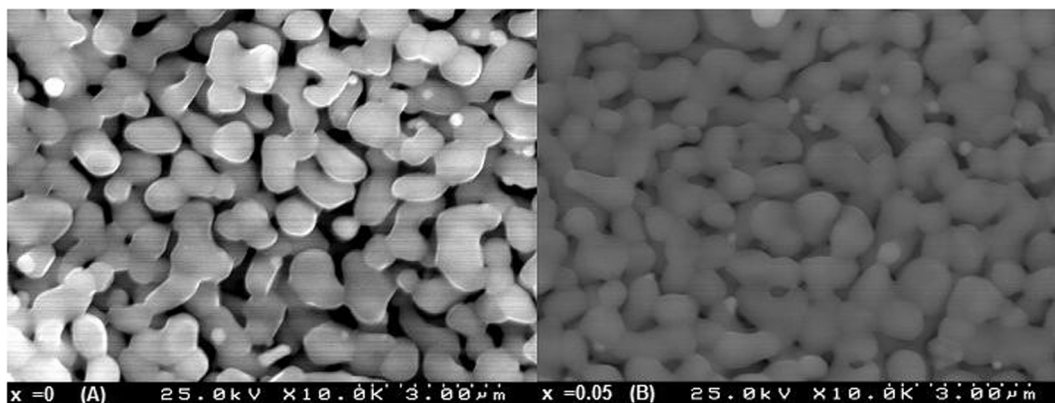


Fig. 11. Microstructural analysis performed on sintered samples of $\text{LiZr}_2(\text{PO}_4)_3$ (A) and $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$ (B) by SEM.

the high (triclinic) and low (monoclinic) temperature phases undergo phase transitions at two different temperatures above room temperature, at around 40–60 °C for the high temperature phase and at around 300 °C for the low temperature phase [10,18].

The XRD spectra presented from Figs. 4–9 show the temperature induced behaviour of $\text{LiZr}_2(\text{PO}_4)_3$, and $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$ ($x = 0.05$) that are inside the limits of solid solution and finally $\text{Li}_{1.10}\text{Zr}_{1.90}\text{B}_{0.10}(\text{PO}_4)_3$ ($x = 0.10$) that is outside the solid solution limit. The observed behaviour is as expected based on the data from the literature [10,18]. One can observe the phase transition from triclinic (PDF#01-074-2562) to rhombohedral (PDF#01-070-6734) for the high temperature phase when heating from 25 °C to 100 °C (Figs. 4–9) and apparently no phase transition from monoclinic (PDF#01-070-5819) for the low temperature phase when heating to 300 °C and above (Figs. 8 and 9), most likely due to the low content of this low temperature phase.

Fig. 10 shows the dilatometry of a green compact rectangular sample of $\text{Li}_{1.025}\text{Zr}_{1.975}\text{B}_{0.025}(\text{PO}_4)_3$. One parameter is densification, Ψ , defined as the change in fractional density due to sintering divided by the fractional density that is needed to attain a pore-free solid [20]:

$$\Psi = \frac{\rho_s - \rho_g}{100 - \rho_g}$$

Sintering shrinkage is coupled to density changes and the elimination of pores. Shrinkage, $\frac{\Delta L}{L_0}$ is the change in compact length divided by the initial dimension. Because of shrinkage, the compact densifies from the fractional green density, ρ_g , to the fractional sintered density, ρ_s , according to the relation [20],

$$\rho_s = \frac{\rho_g}{\left(1 - \frac{\Delta L}{L_0}\right)^3}$$

This is a mass-conservation equation that assumes no mass loss. In reality, powder compacts usually have contaminants and polymers during sintering, so the green density needs to be corrected for such mass loss. High green densities result in high final densities, even with small sintering shrinkages. Densification, final density and shrinkage are related measures of the particle bonding and pore elimination during sintering. Unfortunately, we have not been able to densify the pellet to 80% above in more than 100 attempts. In order to fully understand the cooperative effect of hydraulic pressing and sintering temperature, temperature dependence of shrinkage analysis is needed next. Fig. 11 shows the SEM preformed in fractures of samples of $\text{LiZr}_2(\text{PO}_4)_3$ (A) and $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$ (B) sintered with the selected cycle, with a dwell at 1200 °C for 12 h, both only crystallized in the high temperature triclinic phase. The SEM pictures confirm the low density achieved for

both samples as one can observe large pores along the grain boundary and also a small grain size, especially for the $\text{LiZr}_2(\text{PO}_4)_3$ (A). This along with the failed attempts of densification confirms that more work was needed to achieve dense pellets for future electrical characterization.

4. Conclusions

The method adapted from the recent work by Kim et al. does not offer good reproducibility as we could not achieve phase purity for $\text{LiZr}_2(\text{PO}_4)_3$. It was possible to achieve phase purity for $\text{LiZr}_2(\text{PO}_4)_3$ with the method adapted from Alamo et al. Zr-site is the preferential mechanism for B doping of $\text{LiZr}_2(\text{PO}_4)_3$ and not the P-site. It is possible to obtain phase purity only up to $x = 0.05$, which coincides with the XRD data that after $\text{Li}_{1.075}\text{Zr}_{1.925}\text{B}_{0.075}(\text{PO}_4)_3$ there is no solid solution. Despite the efforts, it was not possible to obtain dense pellets and with physical integrity after sintering, suggesting more work would be needed in order to try further approaches to progress to the next goal of measuring impedance for $\text{LiZr}_2(\text{PO}_4)_3$ and $\text{Li}_{1.05}\text{Zr}_{1.95}\text{B}_{0.05}(\text{PO}_4)_3$.

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References

- [1] Y. Shimonishi et al., A study on lithium/air secondary batteries—stability of the NASICON-type lithium ion conducting solid electrolyte in alkaline aqueous solutions, *J. Power Sources* 196 (2011) 5128–5132.
- [2] Y. Shimonishi et al., A study on lithium/air secondary batteries—stability of NASICON-type glass ceramics in acid solutions, *J. Power Sources* 195 (2010) 6187–6191.
- [3] V. Thangadurai, A.K. Shukla, J. Gopalakrishnan, New lithium-ion conductors based on the NASICON structure, *J. Mater. Chem.* 9 (1999) 739.
- [4] H. Xie, J.B. Goodenough, Y. Li, $\text{Li}_{1.2}\text{Zr}_{1.9}\text{Ca}_{0.1}(\text{PO}_4)_3$, a room-temperature Li-ion solid electrolyte, *J. Power Sources* 196 (2011) 7760–7762.
- [5] A. Ramzy, V. Thangadurai, Tailor-made development of fast Li ion conducting garnet-like solid electrolytes, *ACS Appl. Mater. Interf.* 2 (2010) 385–390.
- [6] A. Clearfield, *Proton Conductors: Solids, Membranes and Gels—Materials and Devices*, Cambridge University Press, Cambridge, 1992.
- [7] A.B. Yaroslavtsev, I.A. Stenina, Complex phosphates with the NASICON structure ($\text{M}_x\text{A}_2(\text{PO}_4)_3$), *Russ. J. Inorg. Chem.* 51 (2006) S97–S116.

- [8] D.A. Woodstock, P. Lightfoot, Comparison of the structural behaviour of the low thermal expansion NZP phases $\text{MTi}_2(\text{PO}_4)_3$ ($\text{M} = \text{Li}, \text{Na}, \text{K}$), *J. Mater. Chem.* 9 (1999) 2907–2911.
- [9] H.Y.-P. Hong, Crystal structures and crystal chemistry in the system $\text{Na}_{1+x}\text{Zr}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$, *Mater. Res. Bull.* 11 (1976) 173–182.
- [10] J.E. Iglesias, C. Pecharroman, Room temperature triclinic modification of NASICON-type $\text{LiZr}_2(\text{PO}_4)_3$, *Solid State Ionics* 112 (1998) 309–318.
- [11] M. Catti, S. Stramare, R. Ibberson, Lithium location in NASICON-type Li conductors by neutron diffraction. I. Triclinic α' - $\text{LiZr}_2(\text{PO}_4)_3$, *Solid State Ionics* 123 (1999) 173–180.
- [12] M. Casciola et al., Preparation, structural characterization and conductivity of $\text{LiZr}_2(\text{PO}_4)_3$, *Solid State Ionics* 26 (1988) 229–235.
- [13] T. Suzuki et al., Stabilization of superionic conduction phase in $\text{Li}_3\text{Sc}_2(\text{PO}_4)_3$, *Solid State Ionics* 104 (1997) 27–33.
- [14] T. Šalkus et al., Fabrication and electrical properties of $\text{Li}_3\text{Sc}_{2-x}\text{B}_x(\text{PO}_4)_3$ superionic materials, *Lith. J. Phys.* 45 (4) (2005) 257–262.
- [15] R.S. Futagami, M. Nishimori, H. Nishizawa, Hydrothermal synthesis and electric conductivity of the NASICON-related solid solution, $\text{Na}_{1+2x}\text{Ti}_2\text{B}_x\text{P}_{3-x}\text{O}_{12}$, *J. Mater. Sci. Lett.* 19 (2000) 405–407.
- [16] J. Alamo, J.L. Rodrigo, Thermal expansion of $\text{LiZr}_2(\text{PO}_4)_3$: water inclusion influence, *Solid State Ionics* 32 (33) (1989) 70–76.
- [17] H.S. Kim, H.S. Ahnn, J.H. Lee, Y.K. Sun, Chong S. Yoon, Electrochemical properties of sol-gel prepared $\text{Li}_2\text{Zr}_x\text{Ti}_{1-x}(\text{PO}_4)_3$ electrodes for lithium secondary batteries, *J. Electrochem. Soc.* 158 (4) (2011) A396–A399.
- [18] F. Suderau, D. Petit, J.P. Boilot, Dimorphism, phase transitions, and transport properties in $\text{LiZr}_2(\text{PO}_4)_3$, *J. Solid State Chem.* 83 (1989) 78–90.
- [19] R.D. Shannon, Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides Data retrieved from < on 20-02-2013 *Acta Crystallogr. A* A32 (5) (1976) 751–767.
- [20] Z. Fang, 1 – Thermodynamics of sintering, in: *Sintering of Advanced Materials*, Ed Woodhead Publishing, 2010, pp. 3–32.